

RESEARCH INTERESTS	Applied spatial statistics with a focus on biological and ecological systems, Bayesian statistics, computational methods, statistics and computing education and pedagogy.	
EDUCATION	University of California, Los Angeles, Department of Statistics Ph.D. in Statistics, 2012 M.S. in Statistics, 2008 Dissertation Topic: Bayesian Methods for Spatial Assignment of Migratory Birds Advisors: Jan de Leeuw and John Novembre California Institute of Technology B.S. in Biology, 2003	
EMPLOYMENT	Associate Professor of the Practice Department of Statistical Science, Duke University	May 2022 - Present
	Assistant Professor of the Practice Department of Statistical Science, Duke University	June 2015 - May 2022
	Lecturer in Statistics and Data Science School of Mathematics, University of Edinburgh	May 2019 - May 2021
	Visiting Assistant Professor / Lecturer Department of Statistical Science, Duke University	January 2012 - May 2015
	Postdoctoral Associate Department of Statistical Science, Duke University	July 2012 - April 2014
	Graduate Student Researcher Novembre Lab, UCLA	September 2010 - December 2011
	Senior Statistical Consultant Statistical Consulting Center, UCLA	March 2009 - December 2011
	Graduate Teaching Assistant Dept. of Ecology and Evolutionary Biology, Dept. of Statistics, UCLA	September 2006 - July 2010
TEACHING	Duke University Sta 30 - Statistics and Quantitative Literacy - Fa 12 Sta 102 - Introductory Biostatistics - Sp 13 , Sp 14 , Fa 14 , Sp 15 , Fa 15 , Sp 16 , Su 16 Sta 111 - Probability and Statistical Inference - Su 14 Sta 112 - Better Living through Data Science - Fa 16 Sta 230 - Probability - Fa 12 , Sp 14 Sta 323 - Statistical Computing - Sp 16 , Sp 17 , Sp 18 , Sp 19 , Sp 22 , Sp 24 , Sp 25 , Sp 26 Sta 344 / 444 / 644 - Spatio-Temporal Modeling - Sp 17 , Sp 18 , Fa 18 , Fa 22 , Fa 23 Sta 523 - Statistical Programming Fa 14 , Fa 15 , Fa 16 , Fa 17 , Fa 18 , Fa 21 , Fa 22 , Fa 23 , Fa 24 , Fa 25 , Fa 26 Sta 663 - Statistical Computing and Computation - Sp 22 , Sp 23 , Sp 25 , Sp 26 Sta 790 - Advanced Statistical Computing - Sp 19	

University of Edinburgh

Math 08068 - Facets of Mathematics - Regression Modeling Theme - Fa 19

Math 11176 - Statistical Programming - [Fa 19](#), [Fa 20](#)

Math 11205 - Machine Learning in Python - Sp 20, Sp 21

ONLINE TEACHING Coursera - [Statistics with R Specialization](#)

[Bayesian Statistics](#)

[Statistics with R Capstone](#)

[Posit Academy](#)

Learn Shiny with R

PUBLICATIONS

Horton, N. J., Alexander, R., Parker, M., Piekut, A., Rundel, C. (2022). *The Growing Importance of Reproducibility and Responsible Workflow in the Data Science and Statistics Curriculum*. Journal of Statistics and Data Science Education, 30(3), 207-208.

Çetinkaya-Rundel, M., Hardin, J., Baumer, B. S., McNamara, A., Horton, N. J., Rundel, C. (2022). *An educator's perspective of the tidyverse*. Technology Innovations in Statistics Education, 14(1).

Poulsen, J., Beirne, C., Rundel, C., Baldino, M., Kim, S., Knorr, J., Minich, T., Jin, L., Núñez, C., Xiao, S., Mbamy, W., Obiang, G., Masseloux, J., Nkoghe, T., Ebanega, M., Clark, C., Fay, M., Morkel, P., Okouyi, J., White, L., Wright, J. (2021). *Long distance seed dispersal by forest elephants*. Frontiers in Ecology And Evolution. 9, 962. [DOI](#).

Beckman M., Çetinkaya-Rundel M., Horton N., Rundel C., Sullivan A., Tackett M. (2020) *Implementing Version Control With Git and GitHub as a Learning Objective in Statistics and Data Science Courses*. Journal of Statistics Education. 29 (Sup 1), 132 - 144. [DOI](#).

Johnson A., Rundel C., Hu J., Ross K., Rossman A. (2020) *Teaching an Undergraduate Course in Bayesian Statistics: A Panel Discussion*. Journal of Statistics Education. 28 (3), 251 - 261. [DOI](#).

Beirne C., Nuñez C., Baldino M., Kim S., Knorr J., Minich T., Jin L., Xiao S., Mbamy W., Obiang G., Masseloux J., Nkoghe T., Ebanega M., Rundel C., Wright J., Poulsen J. (2019) *Estimation of gut passage time of wild, free roaming forest elephants*. Wildlife Biology. 2019 (1).

Cetinkaya-Rundel M., Rundel C.W. (2017) *Infrastructure and tools for teaching computing throughout the statistical curriculum*. The American Statistician. 72 (1), 58 - 65.

Rundel C.W., Schliep E.M., Holland D., Gelfand A. (2015) *A data fusion approach for spatial analysis of speciated PM_{2.5} across time*. Environmetrics. 26 (8), 515 - 525.

Rundel C.W., Wunder M., Alvarado A.H., Ruegg K., Harrigan R., Schuh A., Jeffrey K., Siegel R., DeSante D.F., Smith T.B., Novembre J. (2013) *Novel statistical methods for integrating genetic and stable isotope data to infer individual-level migratory connectivity*. Molecular Ecology. 22 (16), 4163 - 4176.

de Bocanegra H.T., Rostovsteva D., Çetinkaya M., Rundel C.W., Lewis C. (2011). *Quality of reproductive health services to limited English proficient patients*. Journal of Health Care for the Poor and Underserved, 22 (4), 1167 - 1178.

Walker D.W., Muffat J, Rundel C.W., Benzer S. (2006). *Overexpression of a Drosophila Homolog of Apolipoprotein D Leads to Increased Stress Resistance and Extended Lifespan*. Current Biology, 16 (7), 674 - 679.

MAGAZINES	Rundel, C.W., Cetinkaya-Rundel M. (2016) La Quinta is Spanish for next to Denny's, <i>Chance</i> 29 (2), 53 - 57	
	Rundel C.W. (2002) <i>Genes, Aging, and the Future of Longevity</i> <i>Engineering & Science</i> , 65 (4), 36 - 40.	
TALKS & WORKSHOPS	Posit Conf 2025 (Workshop) Shiny for R	September 2025
	useR! 2025 Parsing Quarto and R Markdown documents in R	August 2025
	Datascience in the classroom 2025 (Invited) Reflections on a decade of teaching statistical computing	July 2025
	Posit Conf 2024 (Workshop) Introduction to Shiny for R	August 2024
	JSM 2024 (Invited) Programmatic Manipulation of Quarto Documents for Teaching	August 2024
	Posit Conf 2023 (Workshop) Introduction to Shiny	September 2023
	JSM 2023 (Invited) Quarto and automation in teaching statistical computing	August 2023
	Preparing to Teach 2023 Teaching-focused careers	August 2023
	Toronto Workshop on Reproducibility 2022 Teaching Statistical Computing with Git and GitHub	2022
	ISBA 2022 An overview of Bayesian computational frameworks for teaching	2022
	RStudio Conf 2022 (Workshop) Introduction to Shiny	July 2022
	ISI Short Course 2021 (Workshop) Teaching Data Science	June 2021
	RStudio Global 2021 parsermd - parsing R Markdown for fun and profit	January 2021
	TLMSCO Teaching computing using git and GitHub	September 2020
	JSM 2020 (Invited) Computation Infrastructure for Teaching Bayesian Modeling	August 2020
	Teaching Statistics and Data Science Online (Online Workshop) Workshop 1: Teaching R online with RStudio Cloud Workshop 2: Building interactive tutorials in R Workshop 3: Teaching computing with Git and GitHub	July 2020
	RStudioConf 2020 livecode: broadcast your live coding sessions from and to RStudio	January 2020
	JSM 2019 ghclass: an R package for managing classes with GitHub	August 2019
	JSM 2019 (Workshop) Reproducible Computing	July 2019
	UseR! 2019 ghclass: an R package for managing classes with GitHub	July 2019
	SDSS 2019 (Invited)	May 2019

Using Rocker containers and CI for teaching R-based courses	
ICOTS10 2018 (Workshop)	July 2018
Teaching Data Science, Reproducibly	
ISBA World Meeting 2018 (Short Course)	June 2018
Reproducible Computing	
Joint Statistical Meetings 2017 (Invited)	August 2017
Moving Away from Ad Hoc Statistical Computing Education	
UseR! 2017 (Tutorial)	July 2017
Data Carpentry: Open and Reproducible Research with R	
Joint Statistical Meetings 2016 (Invited)	August 2016
Statistical Computing as an Introduction to Data Science	
UseR! 2016	July 2016
Continuous Integration and Teaching Statistical Computing with R	
Joint Statistical Meetings 2015	August 2015
Teaching statistical computing leveraging the github ecosystem	
UseR! 2015	July 2015
Teaching R using the github ecosystem	
Data Analytics in Business and Social Science Seminar, Duke SSRI	April 2015
Geospatial data and the R ecosystem	
Joint Statistical Meetings 2014	August 2014
A Data Fusion Approach for Space-Time Analysis of Speciated PM _{2.5}	
Duke Dept of Statistical Science Seminar	February 2014
Using GPUs to improve the computational efficiency of Gaussian process models	
Joint Statistical Meetings 2013	August 2013
GPUs, linear algebra, and efficient computing for Gaussian process models	
UseR! 2013	July 2013
Leveraging GPU libraries for efficient computation of Gaussian process models in R	
Joint Statistical Meetings 2012	August 2012
Leveraging GPU Libraries for Efficient Computation of Bayesian Spatial Assignment Models in R	
UseR! 2012	June 2012
rgeos: spatial geometry predicates and topology operations in R	
Joint Statistical Meetings 2011	August 2011
Spatial Models for Bird Origin Assignment Using Genetic and Isotopic Data	

SERVICE	DSS Master's Advisory Committee	Fall 2017 - Spring 2019, Fall 2024 - present
	Information Technology Advisory Council	Fall 2017 - present
	Chair Fall 2024 - present	
	DSS Computing Committee	Fall 2014 - present
	Chair, Spring 2017 - Spring 2019, Fall 2021 - present	
	Guest Associate Editor, JSDSE	Fall 2021
	Special Issue on Teaching Reproducibility and Responsible Workflow	
	ASA DataFest @ Duke	Fall 2011 - Spring 2019
	Co-organizer	

SOFTWARE

[gradermd](#): Shiny apps for gradescope like grading of GitHub based assignments

[md4r](#): R wrapper of the md4c markdown parsing library.

[checklist](#): Tools for automating checking of R projects, with a focus on automated feedback via CI tooling like GitHub actions.

[parsermd](#): Tools for parsing and programmatically interacting with R Markdown documents.

[learnrhash](#): Tools for recording student results for questions and exercises in learnr documents.

[livecode](#): Library for broadcasting source files during live codeing sessions.

[ghclass](#): Library for managing classroom and assignment related tasks on github.

[rgeos](#): R interface to the Geometry Engine, Open Source (GEOS) library.

[isoscatR](#): R package for smoothed and continuous assignment testing (SCAT) of genetic samples

[RcppGP](#): Tools for efficiently working with Gaussian Processes in R / C++

[mapnik](#): [parser](#) and [generator](#) for the [carto](#) map style language.

MEMBERSHIPS

American Statistical Association